

AI-Powered University & Career Guidance Platform for Sri Lankan A/L Students

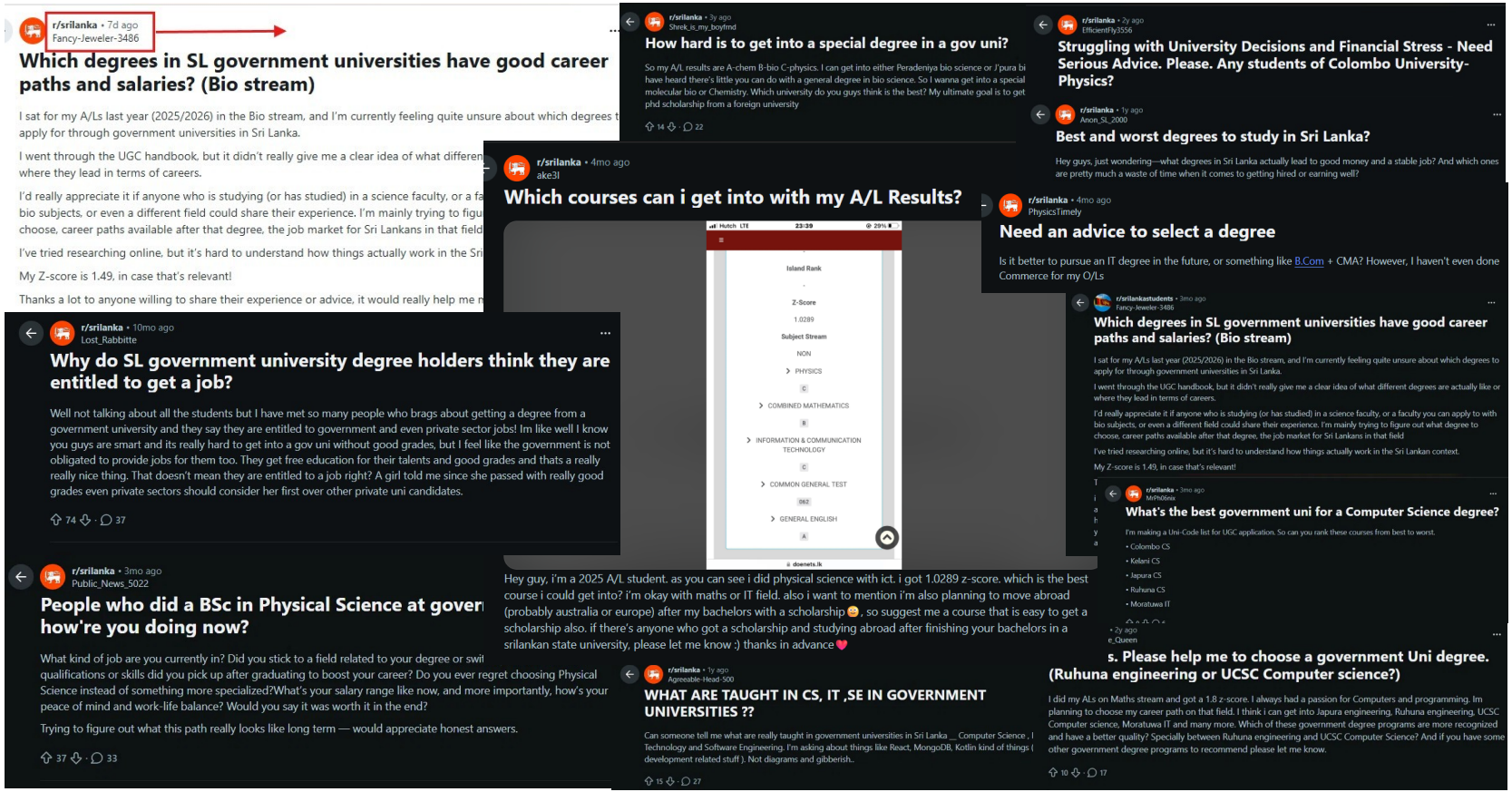
An Intelligent Decision Support System for University Admission and Career Planning

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MSc Artificial Intelligence (Reading)

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Academic Background
🎓 Graduate of the University of Moratuwa (Class of 2023)


r/srilanka • 7d ago
Fancy-Jeweler-3486
Which degrees in SL government universities have good career paths and salaries? (Bio stream)
I sat for my A/Ls last year (2025/2026) in the Bio stream, and I'm currently feeling quite unsure about which degrees t
apply for through government universities in Sri Lanka.
I went through the UGC handbook, but it didn't really give me a clear idea of what differen
where they lead in terms of careers.
I'd really appreciate it if anyone who is studying (or has studied) in a science faculty, or a fa
bio subjects, or even a different field could share their experience. I'm mainly trying to figu
choose, career paths available after that degree, the job market for Sri Lankans in that field
I've tried researching online, but it's hard to understand how things actually work in the Sri
My Z-score is 1.49, in case that's relevant!
Thanks a lot to anyone willing to share their experience or advice, it would really help me
r/srilanka • 3y ago
Shrek_is_my_boyfrnd
How hard is to get into a special degree in a gov uni?
So my A/L results are A-chem B-bio C-physics. I can get into either Peradeniya bio science or J'pura bi
have heard there's little you can do with a general degree in bio science. So I wanna get into a special
molecular bio or Chemistry. Which university do you guys think is the best? My ultimate goal is to get
phd scholarship from a foreign university
14 · 22
r/srilanka • 2y ago
EfficientFly3556
Struggling with University Decisions and Financial Stress - Need Serious Advice. Please. Any students of Colombo University-Physics?
r/srilanka • 1y ago
Anon_SL_2000
Best and worst degrees to study in Sri Lanka?
Hey guys, just wondering—what degrees in Sri Lanka actually lead to good money and a stable job? And which ones are pretty much a waste of time when it comes to getting hired or earning well?
r/srilanka • 4mo ago
ake3l
Which courses can i get into with my A/L Results?
Hey guy, i'm a 2025 A/L student. as you can see i did physical science with ict. i got 1.0289 z-score. which is the best course i could get into? i'm okay with maths or IT field. also i want to mention i'm also planning to move abroad (probably australia or europe) after my bachelors with a scholarship, so suggest me a course that is easy to get a scholarship also. if there's anyone who got a scholarship and studying abroad after finishing your bachelors in a srilankan state university, please let me know :) thanks in advance
Island Rank
Z-Score
1.0289
Subject Stream
NON
PHYSICS
C
COMBINED MATHEMATICS
B
INFORMATION & COMMUNICATION TECHNOLOGY
C
COMMON GENERAL TEST
062
GENERAL ENGLISH
A
r/srilanka • 4mo ago
PhysicsTimely
Need an advice to select a degree
Is it better to pursue an IT degree in the future, or something like B.Com + CMA? However, I haven't even done Commerce for my O/Ls
r/srilanka • 10mo ago
Lost_Rabbitte
Why do SL government university degree holders think they are entitled to get a job?
Well not talking about all the students but I have met so many people who brags about getting a degree from a government university and they say they are entitled to government and even private sector jobs! Im like well I know you guys are smart and its really hard to get into a gov uni without good grades, but I feel like the government is not obligated to provide jobs for them too. They get free education for their talents and good grades and thats a really really nice thing. That doesn't mean they are entitled to a job right? A girl told me since she passed with really good grades even private sectors should consider her first over other private uni candidates.
74 · 37
r/srilankastudents • 3mo ago
Fancy-Jeweler-3486
Which degrees in SL government universities have good career paths and salaries? (Bio stream)
I sat for my A/Ls last year (2025/2026) in the Bio stream, and I'm currently feeling quite unsure about which degrees to apply for through government universities in Sri Lanka.
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I'd really appreciate it if anyone who is studying (or has studied) in a science faculty, or a faculty you can apply to with bio subjects, or even a different field could share their experience. I'm mainly trying to figure out what degree to choose, career paths available after that degree, the job market for Sri Lankans in that field
I've tried researching online, but it's hard to understand how things actually work in the Sri Lankan context.
My Z-score is 1.49, in case that's relevant!
r/srilanka • 3mo ago
MrPh0enix
What's the best government uni for a Computer Science degree?
I'm making a Uni-Code list for UGC application. So can you rank these courses from best to worst.
• Colombo CS
• Kelani CS
• Japura CS
• Ruhuna CS
• Moratuwa IT
r/srilanka • 3mo ago
Public_News_5022
People who did a BSc in Physical Science at gover
how're you doing now?
What kind of job are you currently in? Did you stick to a field related to your degree or swi
qualifications or skills did you pick up after graduating to boost your career? Do you ever regret choosing Physical Science instead of something more specialized?What's your salary range like now, and more importantly, how's your peace of mind and work-life balance? Would you say it was worth it in the end?
Trying to figure out what this path really looks like long term — would appreciate honest answers.
37 · 33
r/srilanka • 1y ago
Agreeable-Head-500
WHAT ARE TAUGHT IN CS, IT ,SE IN GOVERNMENT UNIVERSITIES ??
Can someone tell me what are really taught in government universities in Sri Lanka __ Computer Science , I Technology and Software Engineering. I'm asking about things like React, MongoDB, Kotlin kind of things (development related stuff). Not diagrams and gibberish..
15 · 27
2y ago
e_Queen
s. Please help me to choose a government Uni degree. (Ruhuna engineering or UCSC Computer science?)
I did my ALs on Maths stream and got a 1.8 z-score. I always had a passion for Computers and programming. Im planning to choose my career path on that field. I think i can get into Japura engineering, Ruhuna engineering, UCSC Computer science, Moratuwa IT and many more. Which of these government degree programs are more recognized and have a better quality? Specially between Ruhuna engineering and UCSC Computer Science? And if you have some other government degree programs to recommend please let me know.
10 · 17

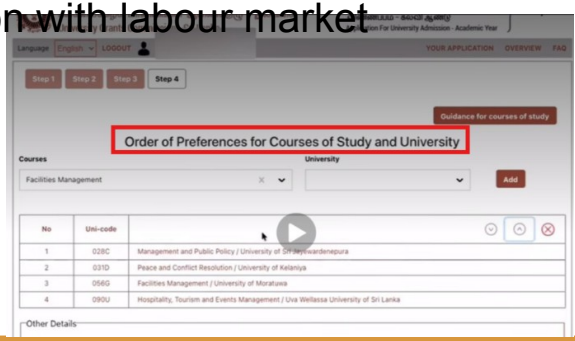
Background & Motivation

Background

- Thousands of Sri Lankan A/L students apply for university admission through the **UGC** each year.
- Students must rank eligible **universities and degree programmes** in order of preference.
- This decision shapes their **higher education and future career**.
- Current resources include the **UGC Admission Handbook** and **Z-score cutoff tables**.

Motivation

- Choosing the right degree programme is a critical career decision.
- Students need reliable, comprehensive information to make informed choices.
- Better guidance can improve employability and align education with labour market needs.



Problem Statement

- The **UGC Admission Handbook** mainly provides:
  - Eligibility requirements
  - Brief programme descriptions
- It provides limited information on:
  - Course curriculum
  - Career opportunities
  - Labour market demand
  - Salary expectations
  - Long-term career growth
- Students often rely on advice from **seniors, graduates, teachers, or family members**, which may be **subjective, biased, or outdated**.

Research Gap

There is **no intelligent decision support system** that integrates:

- Academic eligibility
- Admission probability
- Career insights
- Labour market trends
- Personalized recommendations

"How can Artificial Intelligence be used to provide personalized university and career guidance that enables Sri Lankan A/L students to make informed and confident university preference decisions? "

Main Objectives

Develop an **AI-powered decision support system** that assists Sri Lankan G.C.E. A/L students in selecting and ranking university degree programmes based on their academic performance, interests, and career aspirations.

Specific Objectives

- **Develop an OCR module** to extract student information from A/L result screenshots.
- **Recommend suitable university degree programmes** based on academic eligibility and student preferences.
- **Predict admission probability** using historical admission data and Z-score trends.
- **Provide comprehensive course and career insights**, including curriculum, job opportunities, salary expectations, and labour market demand.
- **Enable students to create and reorder** their university preference list through an interactive drag-and-drop interface.

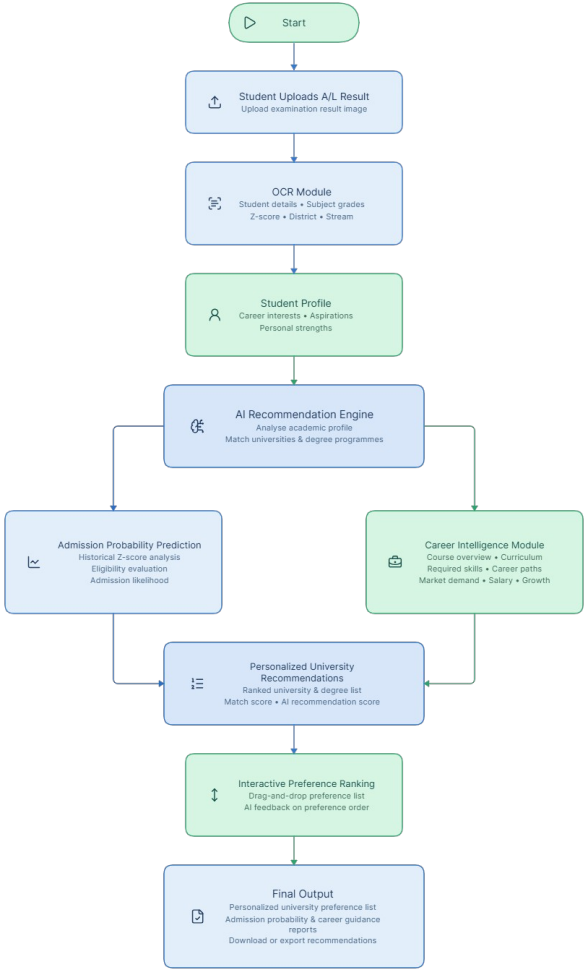
Proposed AI-Based Solution

System Workflow

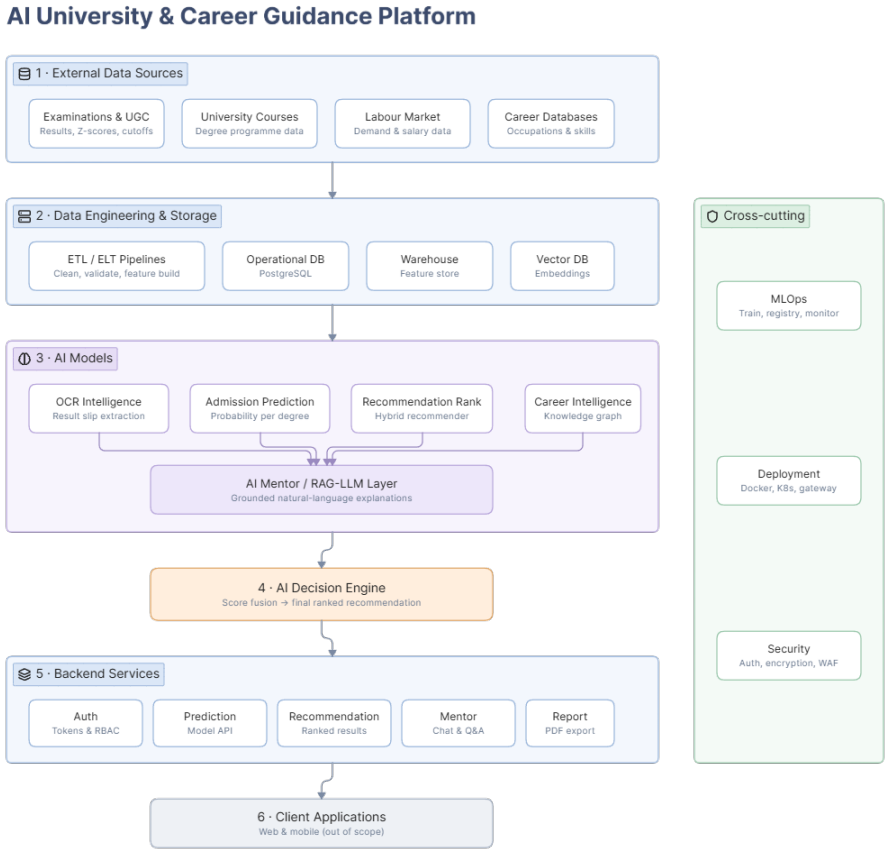
1. Upload A/L Result Screenshot
2. OCR extracts student information
3. Student enters career interests
4. AI analyses academic profile
5. Generate university recommendations
6. Predict admission probability
7. Display career insights
8. Student creates final preference list

Innovation

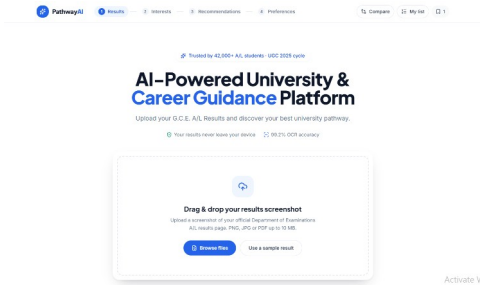
A personalized AI assistant that supports students throughout the university selection process.

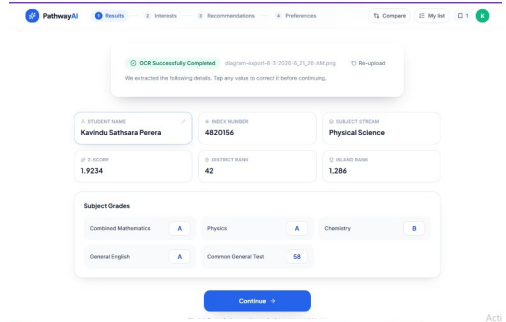


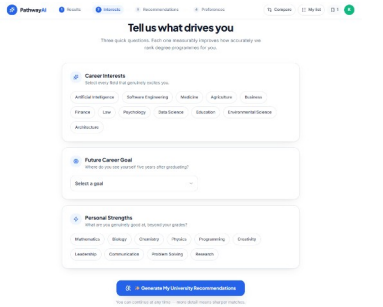
System Architecture

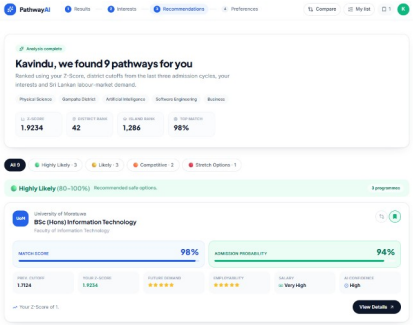


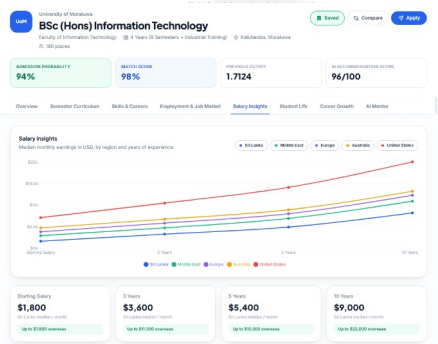
Interactive Prototype

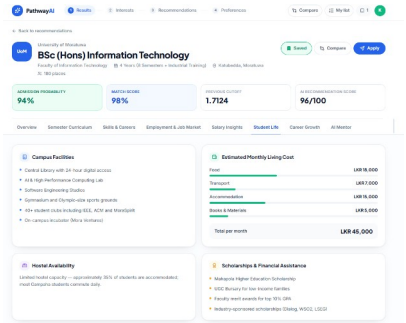












[AI-Powered University & Career Guidance Platform for Sri Lankan A/L Students]

Key Features [1]

AI Recommendation Engine

- Provides personalized university and degree programme recommendations based on student profiles, academic performance, and career goals.

Admission Probability Prediction

- Predicts admission chances and categorizes options into:
  - Safe Choices
  - Moderate Chances
  - Competitive Options

Career Intelligence

- Provides insights into:
  - Potential career opportunities
  - Expected salary trends
  - Labour market demand and future growth

Key Features [2]

Comprehensive Course Information

- Offers detailed degree programme insights, including:
  - Semester-wise curriculum
  - Required skills
  - Student expenses and practical information

Interactive Preference Planning

- Enables students to create and optimize their university preference list through:
  - Drag-and-drop ranking
  - AI-powered feedback and recommendations

User-Friendly Visualization

- Presents insights through modern feature cards, interactive charts, and intuitive visual elements.

Expected Outcomes & Research Contribution

Students

- Make informed education and career decisions
- Increase confidence in university selection
- Reduce uncertainty after A/L results

Universities

- Improve student–programme alignment
- Support better matching between students and suitable degree pathways

Research Contribution

An AI-driven decision support framework integrating academic eligibility, admission probability, and career intelligence into a single platform.

Future Enhancement

- Expand pathways beyond internal university admissions to include:
  - Recognized external degree programmes (e.g., University of Moratuwa BIT, University of Colombo external programmes)
  - Open University programmes
  - Colleges of Education
  - Vocational and professional training pathways
- Integrate real-time labour market trends, salary insights, and career demand analysis.
- Introduce AI career mentor and scholarship recommendation systems.
- Provide Explainable AI-based recommendations for better decision transparency.
- Continuously improve AI models using updated admission and career data.
- Develop a mobile application for wider accessibility.

"To ensure every student can discover a suitable education and career pathway based on their abilities, interests, and aspirations, helping them make informed decisions about their future. "

Conclusion

- Selecting university preferences is a critical decision that shapes students' future academic and career journeys.
- Existing guidance systems provide limited support in connecting academic results with suitable career pathways.
- The proposed AI-powered platform combines personalized recommendations, admission prediction, and career intelligence to provide data-driven guidance.
- The system enables students to make confident, informed decisions about their education and future careers.

"Empowering students to transform academic results into informed career decisions through Artificial Intelligence."

Thank you for your attention.